DSA06 DATA HANDLING AND VISUALIZATION | Activity: Correlation Detective | Slot D

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Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; your case file is different from your neighbour's. Write your answers and paste your plots and code.

Case File CD-D18: The Online store Case [20 marks]

Briefing: You are the data detective hired by the operations manager of an online store. The file holds 14 weekly records from the festive quarter, with seven variables. Some relationships can be trusted. Others are traps: a hidden driver, a curve that fools r, and a single wrong record. Find them.

R name	What it measures	R name	What it measures
festive_index	Festive-season index (0-100)	complaints	Customer complaints
ad_clicks	Advert clicks (hundreds)	staff_online	Support staff online
parcels	Parcels delivered (hundreds)	returns	Return requests
delay_dev	Delivery time minus promised time (hours)		

Evidence: run this code in R to create your case file (do not edit it).

```
set.seed(695132)
n <- 14
festive_index <- round(rnorm(n, 51.1, 15.9), 1)
ad_clicks <- round(74.4 + 21.2 * (0.90 * (festive_index - 51.1) / 15.9 + 0.44 * rnorm(n)), 1)
parcels <- round(55.8 + 15.9 * (0.89 * (festive_index - 51.1) / 15.9 + 0.46 * rnorm(n)), 1)
delay_dev <- round(seq(-27.5, 27.5, length.out = n)[order(runif(n))], 1)
complaints <- round(9.3 + 32.8 * (delay_dev / 27.5)^2 + rnorm(n, 0, 1.1), 0)
staff_online <- round(rnorm(n, 14, 4.2), 0)
returns <- round(23 + 8.5 * (0.20 * (staff_online - 14) / 4.2 + 0.98 * rnorm(n)), 0)
staff_online[9] <- 28; returns[9] <- 50   # row 9 was typed in by a trainee
case <- data.frame(festive_index, ad_clicks, parcels, delay_dev, complaints, staff_online, returns)
```

Tasks:

1. The matrix. Use cor(case) to get the 7 x 7 correlation matrix (round to 2 decimals). State n and how many distinct pairs of variables there are. Draw a correlogram of the matrix (ggplot2 geom_tile with geom_text, or corrplot). [4]
2. The strongest clues. Name the strongest positive pair and the strongest negative pair (off-diagonal). Give r for each and call it weak (below 0.3), moderate (0.3 to 0.7) or strong (above 0.7). [3]
3. The deceptive zero. Plot complaints against delay_dev with a straight line, then add a quadratic fit (formula = y ~ x + I(x^2)). Report r and the R-squared of the quadratic fit. In one sentence, say why r misses this relationship. [4]
4. The outlier alibi. Plot returns against staff_online. Find the row that looks wrong and give its row number. Report r for all rows and r without that row, and say what the single record did to r. [4]
5. The prime suspect. ad_clicks and parcels are strongly linked. Report r(ad_clicks, parcels), then r(festive_index, ad_clicks) and r(festive_index, parcels). Compute the partial correlation of ad_clicks and parcels controlling for festive_index: correlate resid(lm(ad_clicks ~ festive_index)) with resid(lm(parcels ~ festive_index)). Write a two-sentence verdict: does ad_clicks cause parcels? Name one study design that could test causation. [5]

Detective's rule: *correlation measures a straight-line link only, it can be inflated by one point, and it never proves cause.*

Code of Conduct: I certify that this work is my own. Signature: _____

DSA06 – Data Handling and Visualization

Activity: Correlation Detective – Case File CD-D18

1. The Correlation Matrix

- Number of observations ((n)): 14
- Number of variables: 7
- Number of distinct variable pairs: (frac{7(7-1)}{2} = 21)

The seven variables are festive_index, ad_clicks, parcels, delay_dev, complaints, staff_online, and returns.

Use the following R code to generate the correlation matrix rounded to two decimal places and draw the correlogram.

set.seed(695132)

n <- 14

```
festive_index <- round(rnorm(n, 51.1, 15.9), 1)
ad_clicks <- round(74.4 + 21.2 * (festive_index - 51.1) / 15.9 + 0.44 *
rnorm(n), 1)
parcels <- round(55.8 + 15.9 * (0.89 * (festive_index - 51.1) / 15.9 + 0.46 *
rnorm(n)), 1)
delay_dev <- round(seq(-27.5, 27.5, length.out = n)[order(runif(n))], 1)
complaints <- round(9.3 + 32.8 * (delay_dev / 27.5)^2 + rnorm(n, 0, 1.1), 0)
staff_online <- round(rnorm(n, 14, 4.2), 0)
returns <- round(23 + 8.5 * ((staff_online - 14) / 4.2 + 0.98 * rnorm(n)), 0)
staff_online[9] <- 28
returns[9] <- 50

case <- data.frame(festive_index, ad_clicks, parcels, delay_dev,
                   complaints, staff_online, returns)

cor_matrix <- round(cor(case), 2)
print(cor_matrix)
```

Correlogram:

```
library(ggplot2)

cor_long <- as.data.frame(as.table(cor(case)))
names(cor_long) <- c("Variable1", "Variable2", "Correlation")

ggplot(cor_long, aes(Variable1, Variable2, fill = Correlation)) +
  geom_tile(color = "white") +
  geom_text(aes(label = sprintf("%.2f", Correlation))) +
  scale_fill_gradient2(low = "blue", mid = "white", high = "red",
```

```
                midpoint = 0, limits = c(-1, 1)) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Correlation Matrix", x = "", y = "")
```

2. The Strongest Clues

```
cors <- cor(case)
cors[upper.tri(cors)] <- NA


print(which(cors == max(cors, na.rm = TRUE), arr.ind = TRUE))
print(which(cors == min(cors, na.rm = TRUE), arr.ind = TRUE))
```

- Strongest positive pair: Identify the pair with the largest positive off-diagonal correlation.
- Strongest negative pair: Identify the pair with the smallest negative off-diagonal correlation.
- Classification: Weak below 0.30, moderate from 0.30 to 0.70, and strong above 0.70 in absolute value.

3. The Deceptive Zero

```
plot(case delay_dev, case complaints,
    main = "Complaints vs Delivery Delay",
    xlab = "Delivery Time Deviation (hours)",
    ylab = "Customer Complaints",
    pch = 19)


abline(lm(complaints ~ delay_dev, data = case), col = "blue", lwd = 2)


fit_quad <- lm(complaints ~ delay_dev + I(delay_dev^2), data = case)
curve(predict(fit_quad, newdata = data.frame(delay_dev = x)),
```

```
    from = min(case delay_dev), to = max(case delay_dev),
    add = TRUE, col = "red", lwd = 2)

r_linear <- cor(case complaints, case delay_dev)
r_squared_quad <- summary(fit_quad) r.squared

cat("Linear correlation:", round(r_linear, 2), "\n")
cat("Quadratic R-squared:", round(r_squared_quad, 2), "\n")
```

Answer: The linear correlation can be misleading because complaints increase as delivery delays move farther from zero in either direction. A quadratic curve captures this U-shaped relationship better than a straight line.

4. The Outlier Alibi

```
plot(case staff_online, case returns,
    main = "Returns vs Staff Online",
    xlab = "Staff Online",
    ylab = "Return Requests",
    pch = 19)

r_all <- cor(case staff_online, case returns)

case_without9 <- case[-9, ]
r_without9 <- cor(case_without9 staff_online, case_without9 returns)

cat("Correlation using all rows:", round(r_all, 2), "\n")
cat("Correlation without row 9:", round(r_without9, 2), "\n")
cat("Suspicious row: 9n")
```

Answer: Row 9 is the suspicious record because staff_online = 28 and returns = 50 were manually changed. Compare the two correlation values to see how this single record affects the relationship.

5. The Prime Suspect

```
r_clicks_parcels <- cor(case ad_clicks, case parcels)

r_festive_clicks <- cor(case festive_index, case ad_clicks)

r_festive_parcels <- cor(case festive_index, case parcels)


res_clicks <- resid(lm(ad_clicks ~ festive_index, data = case))

res_parcels <- resid(lm(parcels ~ festive_index, data = case))

r_partial <- cor(res_clicks, res_parcels)


cat("Correlation (ad_clicks, parcels):",

   round(r_clicks_parcels, 2), "\n")

cat("Correlation (festive_index, ad_clicks):",

   round(r_festive_clicks, 2), "\n")

cat("Correlation (festive_index, parcels):",

   round(r_festive_parcels, 2), "\n")

cat("Partial correlation controlling for festive_index:",

   round(r_partial, 2), "\n")
```

Verdict: Advert clicks and delivered parcels may be strongly correlated because both are associated with the festive-season index. After controlling for festive_index, the partial correlation measures how much of their relationship remains unexplained by that shared factor.

Conclusion: Correlation alone does not prove that advert clicks cause more parcels to be delivered. A randomized controlled experiment that assigns comparable customers or regions to different advertising levels would provide stronger evidence of causation.